Some Basic Probability

Summary Continuous RV

- Expected Values

Standard Normal Random Variable

- Definition
- Probability Calculations

Normal Random Variable

- Relation to Standard Normal Random Variable
- Probability Calculations
- Testing if Data are Normally Distributed

BME 423 Discussion - Week 3

Outline

Last Week

I. Recap - Week 2

II. Plotting (Week 2 Cont.)

Descriptive Statistics

I. Measures of Central Tendency: Mean, Median, Trimmed Mean, Mode

II. Measures of Variability: Range, Interquartile Range, Variance, Standard Deviation

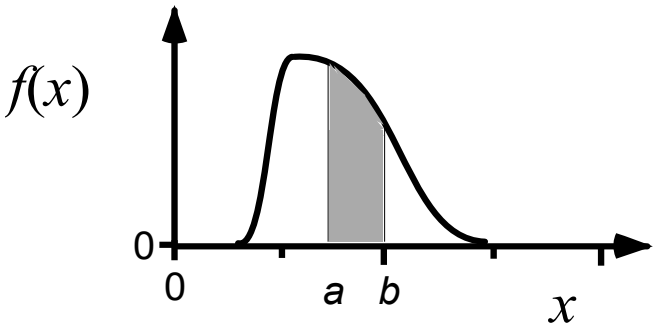
III. Skewness, Kurtosis

IV. Descriptive Statistics Functions
- summary()
- describe()
- describeBy()

Continuous Random Variable *X*

$f(x)$ Probability Density Function (p.d.f.)

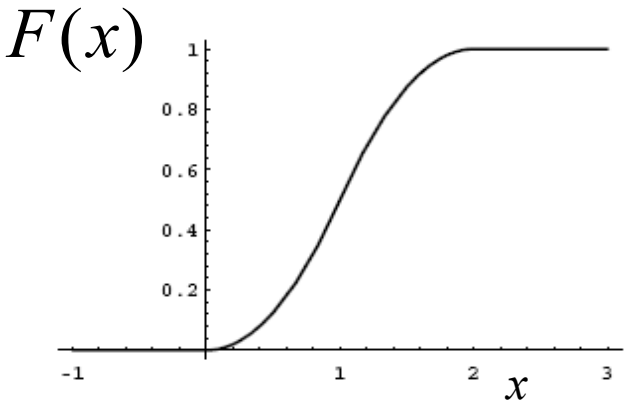
$$\text{Prob}(a < X \le b) = \int_a^b f(x)dx$$



$F(x)$ Distribution Function (d.f.)

$$F(x) = \text{Prob}(X \le x) = \Pr(X \le x)$$

$$F(x) = \int_{-\infty}^{x} f(u) \cdot du$$



Expected Value

Let X denote a continuous RV with $X \sim f(x)$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

Define a function of X as $h(X)$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

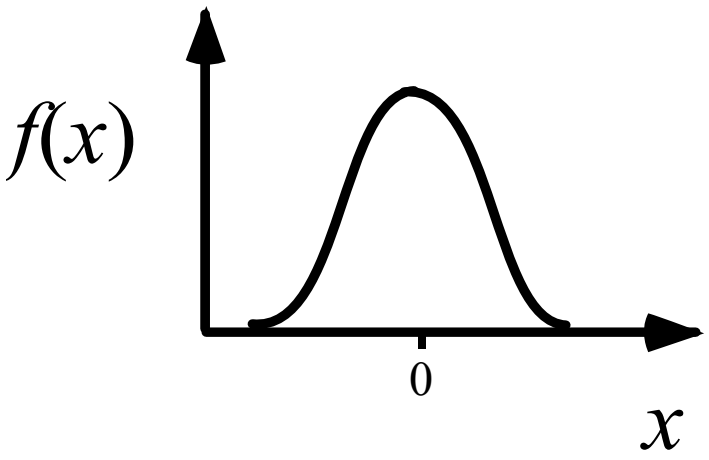
Other Relationships Involving d.f.

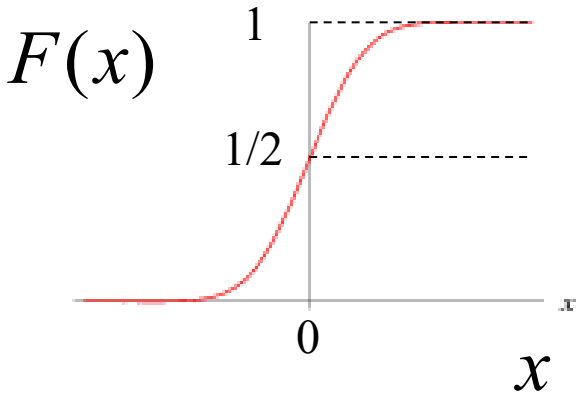
Recall: $F(x) = \Pr(X \le x)$ $\qquad F(x) = \int_{-\infty}^{x} f(u) \cdot du$

$$\Pr(X > x) = 1 - F(x)$$

$$\Pr(x_1 < X \le x_2) = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x)\,dx$$

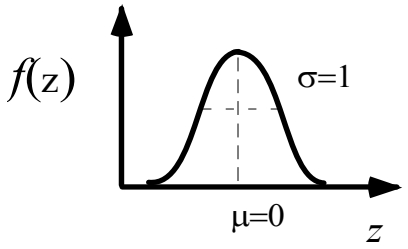
Figures to help illustrate these ideas





Standard Normal RV *Z*

$$Z \sim N(0, 1^2)$$



$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-(z)^2}{2}\right)$$

p.d.f.

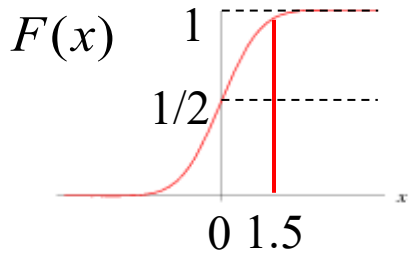
d.f. → $$F(z) = \int_{-\infty}^{z} \frac{1}{(2\pi)^{1/2}} \exp\left(-\frac{1}{2}u^2\right) du$$

Table of the Standard Normal Distribution Function

z	$F(z)$
0.00	0.5000
0.01	0.5040
0.02	0.5080
0.03	0.5120
0.04	0.5160
0.05	0.5199

z	$F(z)$
3.55	0.9998
3.60	0.9998
3.65	0.9999
3.70	0.9999
3.75	0.9999
3.80	0.9999
3.85	0.9999
3.90	1.0000

Examples: $Z \sim N(0, 1^2)$



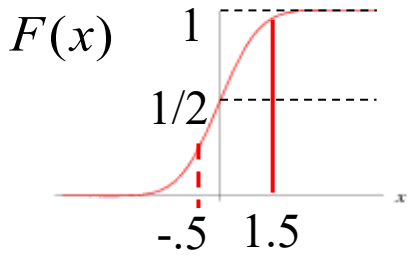
$$\Pr(Z < 1.5) = F(1.5) = 0.9332$$

$$\Pr(Z > 0.5) = 1 - F(0.5) = 1 - 0.6915 = 0.3085$$

$$\Pr(0.2 < Z \le 0.4) = F(0.4) - F(0.2) = 0.6554 - 0.5793 = 0.0761$$

$$\Pr(Z < -2.0) = \Pr(Z > 2.0) = 1 - F(2.0) = 1 - 0.9773 = 0.0227$$

$$\Pr(-0.5 < Z \le 1.5) = F(1.5) - F(-0.5)$$
$$= F(1.5) - \big(1 - F(0.5)\big) = 0.6247$$



$$\Pr(-1 < Z) = 1 - \Pr(Z \le -1) = 1 - \big(1 - F(1)\big) = F(1) = \Pr(Z \le 1) = 0.8413$$

Exercise – Finding Probabilities

- If Z is a Standard Normal random variable, calculate the following probabilities using the Standard Normal Distribution Function table.
- You should also check your answers using the **`pnorm()`** function in R: `pnorm(q, mean = 0, sd = 1)`, where `q` is value of z; function returns the value of $F(z)$.

$$\Pr(Z \leq 1.37)$$

$$\Pr(0 < Z \leq 2.17)$$

$$\Pr(1.5 \leq Z)$$

$$\Pr(|Z| \leq 2.5)$$

$$\Pr(-1 < Z \leq 1)$$

$$\Pr(-2 < Z \leq 2)$$

$$\Pr(-3 < Z \leq 3)$$

Exercise – Finding Quantiles

A quantile is the value of the random variable associated with a given/specific probability value.

- Find the value of the constant c (quantile) that makes the probability statement correct for a Standard Normal random variable Z.
- You will need to use the Standard Normal Distribution Function table.
- You should also check your answers using the **`qnorm()`** function in R: `qnorm(p, mean = 0, sd = 1)`, where `p` is $F(c)$; function returns the value of c.

$$\Pr(Z \leq c) = 0.937$$

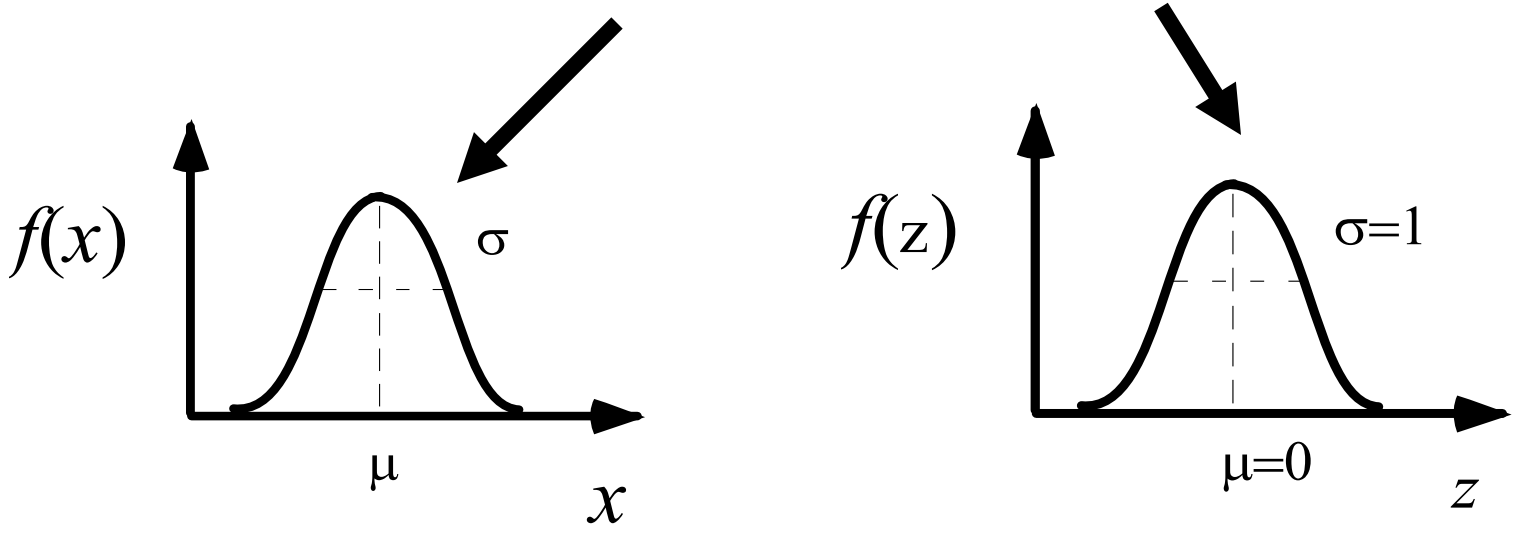
$$\Pr(Z \leq c) = 0.975$$

$$\Pr(c \leq Z) = 0.121$$

$$\Pr(-c < Z \leq c) = 0.95$$

Normal RV and Standard Normal RV

Relationship between $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0, 1^2)$

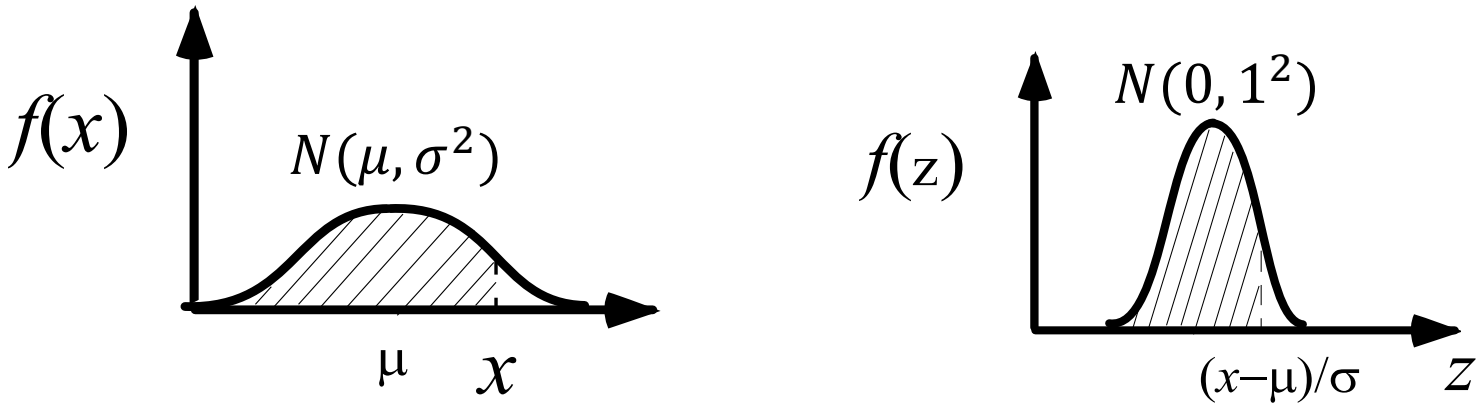


Define $x = \mu + \sigma z$

therefore $z = \dfrac{x - \mu}{\sigma}$

The following can be shown:

$$\Pr(X \le x) = \Pr\left(Z \le \frac{x-\mu}{\sigma}\right)$$



Therefore, the following are also true (for example):

$$\Pr(X > x) = 1 - \Pr(X \le x) = 1 - \Pr\left(Z \le \frac{x-\mu}{\sigma}\right)$$

$$\Pr(a < X \le b) = \Pr(X \le b) - \Pr(X \le a) = \Pr\left(Z \le \frac{b-\mu}{\sigma}\right) - \Pr\left(Z \le \frac{a-\mu}{\sigma}\right)$$

Probability Calculations

Let RV X represent arterial P_{O_2} (mmHg) where $X \sim N(90, 7^2)$

$\Pr(X < 110 \text{ mmHg}) = ?$ $\Pr(80 < X < 110 \text{ mmHg}) = ?$

Recall $x = \mu + \sigma z$ $z = \dfrac{x - \mu}{\sigma}$

$$\Pr(X < x) = \Pr\left(Z < \frac{x - \mu}{\sigma}\right)$$

$$\Pr(X < 110) = \Pr\left(Z < \frac{110 - 90}{7}\right) = \Pr(Z < 2.86)$$

$$= F(2.86) = 0.9979$$

Probability Calculations

If X is a Normal random variable calculate the following probabilities

$$\Pr(\mu - \sigma < X \leq \mu + \sigma)$$

$$\Pr(\mu - 2\sigma < X \leq \mu + 2\sigma)$$

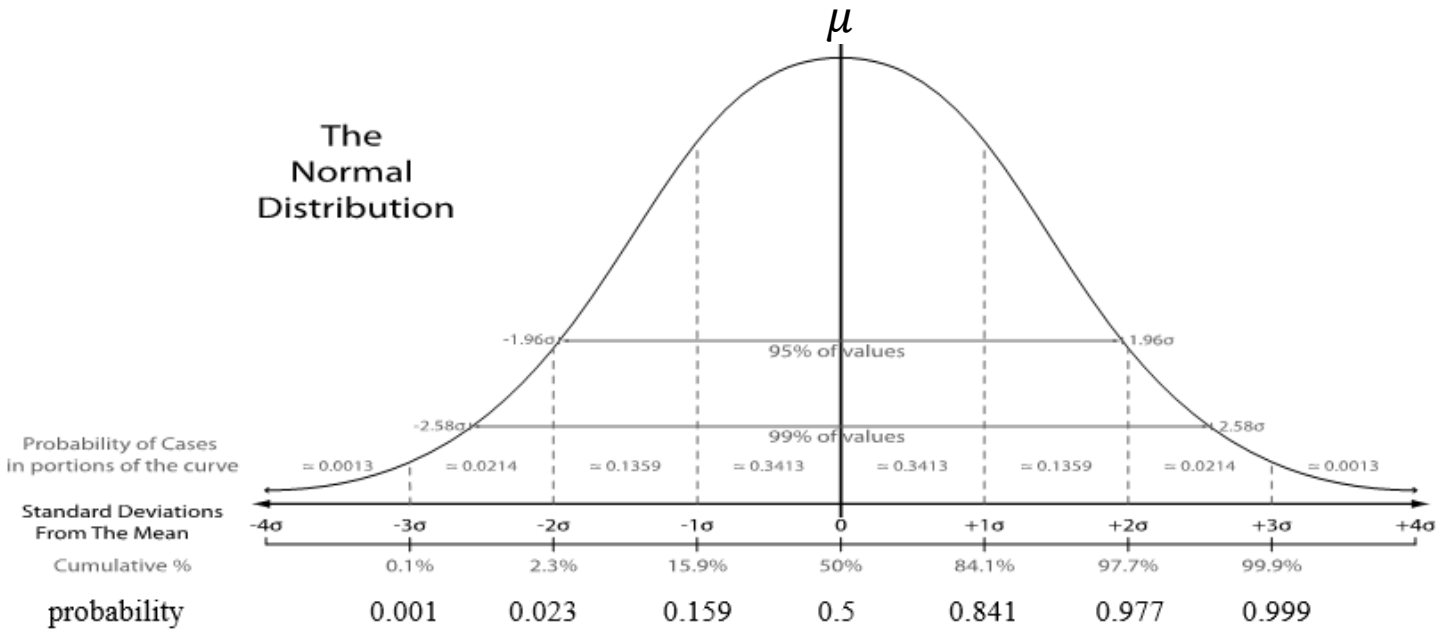
$$\Pr(\mu - 3\sigma < X \leq \mu + 3\sigma)$$

Probability Calculations

The amount of an impurity in a batch of a chemical product is a Normally distributed random variable X with mean of 4.0 grams and standard deviation of 1.5 grams

If 50 batches are independently prepared, what is the probability that the sample mean amount of impurity, $\bar{x}$, is between 3.5 and 3.8 grams?

Testing if Data are Normally Distributed



We previously introduced a qualitative assessment of whether data follow a Normal distribution that involved comparing the percentage of sample data in a range to the corresponding values we would expect for Normally distributed data (e.g., ~68% between $\bar{x} - s$ & $\bar{x} + s$, ~2.5% from $-\infty$ to $\bar{x} - 2s$, etc.)

This idea will now be generalized using the p.d.f. for a Normal distribution.

Quantile-Quantile Plot for Testing if Data are Normally Distributed

A general Quantile-Quantile Plot (Q-Q plot) compares the quantiles of two sets of data.

We will use a version of this idea (a Normal Quantile Plot) for testing if data follow a Normal distribution:

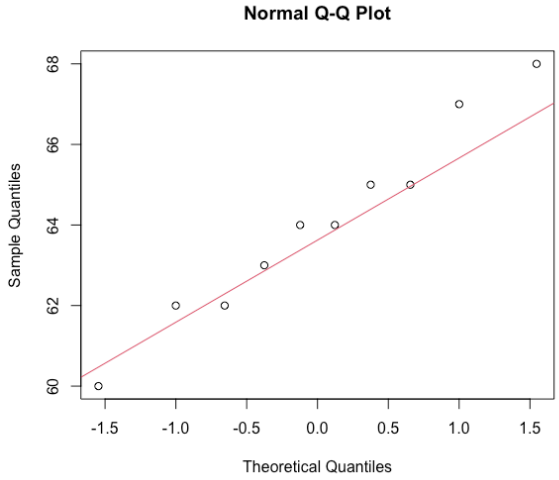
- It plots the sample quantile of each data point versus the expected Z value for that data point.
- If the sample quantiles match their theoretical (Normal) quantiles the graph will be a straight line.
- Even when sample data *are* drawn from a Normal distribution, the plot will have some deviations from a straight line due to the effects of sampling.

We will use the R function **`qqnorm()`**, which performs the needed quantile calculations and does the plotting.

Testing if Data are Normally Distributed using **`qqnorm()`** function

Displayed below is the `qqnorm()` analysis of 10 data points x_i

```
x <- c(60, 62, 62, 63, 64, 64, 65, 65, 67, 68)
qqnorm(x)
qqline(x, col=2)
```



Might conclude that the result follows a straight line, indicating the data come from a Normal distribution.

In contrast, the examples below demonstrate non-Normal distributions (from A. Tanbakuchi)

